

Student 3 **DANIMIZA V** 192421363RC

10 / 10 submitted

Q1. Peak Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
a = [1,3,20,4,1,0]
n = len(a)
for i in range(n):
    if(i == 0 or a[i] >= a[i-1]) and (i == n - 1 or a[i] >= a[i+1]):
        print(a[i])
        break
```

OUTPUT

20

Q2. Second Largest and Second Smallest - Tuple

Score: 0.00 TC: 0/1 passed

CODE

```
lst = [10,20,5,40,15,30]
t = tuple(lst)
s = sorted(t)
print(s[1])
print(s[-2])
```

OUTPUT

10
30**Q3. Frequency of Each Element - List**

Score: 0.00 TC: 0/1 passed

CODE

```
a = [1,2,3,4,1,5,2]
visited = []
for i in a:
    if i not in visited:
        print(i, "=", a.count(i))
        visited.append(i)
```

OUTPUT

1 = 2
2 = 2
3 = 1
4 = 1
5 = 1**Q4. Missing Number - Tuple**

Score: 10.00 TC: 1/1 passed

CODE

```
t = (1,2,3,5,6)
n = max(t)
total = n * (n+1) // 2
missing = total - sum(t)
print(missing)
```

OUTPUT

4

Q5. Majority Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
def majority_element(lst):
    n = len(lst)
    for i in lst:
        if lst.count(i)>n//2:
            return i
    return -1

lst = [2,2,1,2,3,2,2]
print(majority_element(lst))
```

OUTPUT

2

Q6. Sum & Average - Tuple

Score: 0.00 TC: 0/1 passed

CODE

```
t = (10,20,30,40,50)
total = sum(t)
average = total / len(t)
print(total)
print(average)
```

OUTPUT150
30.0**Q7. Element Exists - Tuple**

Score: 0.00 TC: 0/2 passed

CODE

```
t=(10,20,30,40,50)
x = 30
if x in t:
    print("element found")
else:
    print("element not found")
x = 25
if x in t:
    print("element found")
else:
    print("element not found")
```

OUTPUTelement found
element not found**Q8. Maximum Element - Tuple**

Score: 10.00 TC: 1/1 passed

CODE

```
t = (12,45,8,67,23)
print(max(t))
```

INPUT

12 45 8 67 23

OUTPUT

67

SIMATS Engineering • CSE • CSA0802 — Python Programming • 31-07-2026 • Category: Class Work (30.7.26) • Student Submissions Report

Q9. Lucky Number

Score: 0.00 TC: 0/2 passed

CODE

```
def digit_sum(n):
    if n < 10:
        return n
    return n % 10 + digit_sum(n // 10)
def lucky(n):
    if n < 10:
        return n == 7
    return lucky(digit_sum(n))
n1 = 34
n2 = 123
if lucky(n1):
    print("Lucky Number")
else:
    print("Not a Lucky Number")
if lucky(n2):
    print("Lucky Number")
else:
    print("Not a Lucky Number")
```

OUTPUT

Lucky Number
Not a Lucky Number

Q10. Remove Duplicate Elements - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
t = (10,20,30,20,40,10,50)
t = tuple(set(t))
print(t)
```

OUTPUT

(40, 10, 50, 20, 30)